

CORE — the four tweens

```
gsap.to('.box', {x:300, rotation:360, duration:1})
gsap.from('.el', {y:60, opacity:0}) // animate IN
gsap.fromTo('.el', {scaleX:0},
             {scaleX:1, transformOrigin:'left'})
gsap.set('.el', {autoAlpha:0}) // instant, no anim
```

autoAlpha = opacity + visibility. Prefer x/y/scale/rotation (GPU).

COMMON PROPERTIES

x, y, scale, rotation, skew, opacity, autoAlpha
duration, delay, repeat, repeat:-1, yoyo:true
ease, stagger, transformOrigin
backgroundColor, color, '--myVar' (CSS vars)
onComplete, onUpdate, onStart (callbacks)

EASES — the secret sauce

```
ease:'power2.out' // fast start, soft end (UI)
ease:'power4.inOut' // strong both ends
ease:'back.out(1.7)' // overshoots
ease:'elastic.out(1,0.3)' // springy
ease:'bounce.out' 'sine' 'expo' 'none'
```

power2.out is the safe default for buttons, cards, menus.

STAGGER — many elements

```
gsap.from('.item', {opacity:0, y:40,
                    stagger:0.08}) // 0.08s apart

stagger:{each:0.05, from:'center', grid:'auto'}
```

TIMELINE — sequencing

```
const tl = gsap.timeline({
  defaults:{duration:0.5, ease:'power2.out'}})

tl.from('.logo', {y:-40, opacity:0})
  .from('.title', {x:-60, opacity:0}, '<') // together
  .from('.text', {opacity:0}, '-=0.3') // overlap
  .from('.cta', {scale:0.8}, '+=0.2') // gap
  .addLabel('mid') // label

// control: play() pause() reverse()
//           timeScale(2) seek('mid')
```

Position param: '<' with prev, '-=x' overlap, '+=x' gap, 'label'.

SCROLLTRIGGER — scroll magic

```
gsap.registerPlugin(ScrollTrigger)

gsap.from('.section', {opacity:0, y:80,
  scrollTrigger:{
    trigger:".section",
    start:"top 80%", // el top hits 80% vh
    end:"bottom 20%",
    scrub:true, // tie to scroll
    pin:true, // fix while scrolling
    toggleActions:"play none none reverse",
    markers:true // debug start/end
  }})
```

scrub:1 = smooth catch-up. end:'+=1000' = pin for 1000px.

SPLITTEXT — animate text

```
import {SplitText} from 'gsap/SplitText'
gsap.registerPlugin(SplitText)

const s = new SplitText('.headline',
                        {type:'chars,words,lines'})
gsap.from(s.chars, {y:40, opacity:0,
  stagger:0.03, ease:'back.out(2)'})
```

Now free & 50% smaller. Revert with s.revert().

SVG PLUGINS

```
// DrawSVG — draw a path
gsap.from('.path', {drawSVG:'0%', duration:2})

// MorphSVG — shape to shape
gsap.to('#start', {morphSVG:'#end'})

// MotionPath — move along a path
gsap.to('.rocket', {motionPath:{
  path:'#route', align:'#route', autoRotate:true}})
```

FLIP — layout transitions

```
import {Flip} from 'gsap/Flip'
gsap.registerPlugin(Flip)

const st = Flip.getState('.item') // 1. snapshot
container.appendChild(item) // 2. move in DOM
Flip.from(st, {duration:0.6, // 3. animate diff
  ease:'power2.inOut'})
```

Great for shared-element transitions & sortable lists.

REACT — useGSAP

```
import {useGSAP} from '@gsap/react'

useGSAP(() => {
  gsap.from('.title', {y:50, opacity:0})
}, {scope: containerRef}) // auto cleanup
```

scope limits selectors; cleanup runs on unmount (StrictMode-safe).

PERFORMANCE & A11Y

```
// animate transform/opacity, not width/top
// responsive, auto-reverting:
gsap.matchMedia().add('(min-width:768px)', () => {
  gsap.to('.x', {x:200})
})
// honor reduced motion:
const r = matchMedia(
  '(prefers-reduced-motion:reduce)').matches
if (!r) gsap.from('.x', {y:40, opacity:0})
```

Clean up: useGSAP / tween.kill() / ScrollTrigger.kill().